CONSTRAINED FERMI SYSTEMS AND VIRASORO ALGEBRAS

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We study constrained two-dimensional fermion models in the path-integral framework. We consider constraints on the fermion currents and on the energy–momentum tensor. We discuss how these constraints determine the structure of the Virasoro algebras.

Fermi field theories in $(1+1)$ dimensions play a crucial role in our understanding of conformal field theories (CFTs) Onsager's [1] initial solution of the two-dimensional Ising model was later shown to be simply explained in terms of a single Majorana fermion which becomes massless at the critical point [2]. Very recently much progress has been made in understanding and classifying all two-dimensional fixed points, i.e. all two-dimensional CFTs [3–5]. In particular it was shown by Witten [6] and by Polyakov and Wiegmann [7] that certain two-dimensional Fermi field theories are equivalent to certain Wess–Zumino σ-models at the conformally invariant point. It was later suggested by Goddard, Kent and Olive [8] (GKO) and by Kent [9] that the principal series of Friedan, Qiu and Shenker [4] (FQS) could be related to constrained Fermi CFTs. The algebraic approach of GKO yields an explicit construction of the FQS principal series in terms of *differences* of (properly normalized) Virasoro generators of appropriately chosen Wess–Zumino–Witten models. Their construction suggests that constrained Fermi field theories could be used to represent the FQS series with $C<1$. In this work we explore these ideas by constraining otherwise free Fermi field theories using a chiral gauge field, i.e. by demanding that the generators of a chiral group annihilate the vacuum.

In this paper we reconsider constrained Fermi CFTs from the path-integral formulation. We explicitly construct the Wess–Zumino (WZ) action and calculate the central charges for both the Kac–Moody and Virasoro algebras. We show that simply constraining otherwise free fermions by quotienting (i.e. gauging) a subgroup H of the gauge group does not yield theories in the FQS principal series. We show that the result is a WZ model on a group H at some level k which depends on the representation content of the fermions. These theories are automatically unitary provided k is an integer (as we show it is the case). Thus, in order to have a representation of the FQS series, it may be necessary to venture into generically non-unitary theories such as theories with ghosts and then find adequate consistency conditions. We also consider a free fermion theory constraining a piece of the energy–momentum tensor. We show that this is not equivalent to our first construction but it is the same as subtracting two Virasoro algebras [8].

Consider first a theory of left-handed fermions with N_c colors and N_f flavors. We shall impose the constraint that the allowed states in the Hilbert space be *color singlets*. Thus, the left-handed currents $J_-^a(x)$ $(a=1, 2, ..., N_c^2-1)$ must annihilate the allowed states $\{|f\rangle\}$:

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$$J^a_-(x)\,|\mathrm{f}\rangle=0\,. \tag{1}$$

The generating functional is

$$Z=\int \mathrm{D}\psi_\mathrm{L}^\dagger \mathrm{D}\psi_\mathrm{L} \exp\left(\mathrm{i}\int \psi_\mathrm{L}^{if\dagger}\mathrm{i}\partial_+\psi_\mathrm{L}^{if}\,\mathrm{d}^2x\right) \times\prod_a \delta(J^a_-(x))\,, \tag{2}$$

with $i=1, 2, ..., N_\mathrm{c}$, $f=1, 2, ..., N_\mathrm{f}$, $a=1, 2, ..., N_\mathrm{c}^2-1$.

The constraint can be handled by means of a Lagrange multiplier field $A_+(x_+)$ taking values in the Lie algebra of $\mathrm{SU}(N_\mathrm{c})$:

$$Z=\int \mathrm{D}\psi_\mathrm{L}^\dagger\, \mathrm{D}\psi_\mathrm{L}\, \mathrm{D}A_+ \exp\left(\mathrm{i}\int \psi_\mathrm{L}^{if\dagger}(\mathrm{i}\partial_+ +A_+)\psi_\mathrm{L}^{if}\,\mathrm{d}^2x\right)\,. \tag{3}$$

This amounts to calculating a left-handed fermion determinant. This is not a well-posed problem since the Dirac operator in eq. (3) does not have a well-defined eigenvalue problem. Following Alvarez-Gaumé and Ginsparg [10] we define the left-handed functional integral by adding right-handed decoupled fermion fields. Thus, up to a normalization constant, we have

$$Z=N\int \mathrm{D}\bar\psi\, \mathrm{D}\psi\, \mathrm{D}A_+ \exp\left(\mathrm{i}\int \bar\psi(\mathrm{i}\partial\!\!\!/ +\gamma_-A_+)\psi\,\mathrm{d}^2x\right)\,, \tag{4}$$

and one has to evaluate the determinant of a Dirac fermion in the light-cone gauge. It was first computed by Polyakov and Wiegmann [7] and the result is

$$\det{}_\mathrm{L}(\mathrm{i}\partial_+ +A_+)=\exp[\mathrm{i}S_\mathrm{WZ}(g)]\,, \tag{5}$$

where $g\in\mathrm{SU}(N_\mathrm{c})$ and is defined in terms of A_+:

$$A_+=-\mathrm{i}(g^{-1}\partial_+g)\,. \tag{6}$$

Notice that since A_+ is only function of x_+, $g=A(x_-)B(x_+)$. The effective action $S_\mathrm{WZ}(g)$ is the Wess–Zumino action [6]. By taking into account the flavor degeneracy we can write

$$Z=\int \mathrm{D}g\, J(g)\, \exp[\mathrm{i}N_\mathrm{f}S_\mathrm{WZ}(g)]\,, \tag{7}$$

where $J(g)$ is the Faddeev–Popov determinant for the light-cone gauge,

$$\mathrm{D}A_+=J(g)\mathrm{D}g\,. \tag{8}$$

The jacobian equals [11]

$$J(g)=\exp[\mathrm{i}2N_\mathrm{c}S_\mathrm{WZ}(g)]\,, \tag{9}$$

which generalizes the SU(2) result of Polyakov and Wiegmann [7]. Thus we find that an $\mathrm{SU}(N_\mathrm{c}N_\mathrm{f})$ theory [#1] of fermions constrained down to $\mathrm{SU}(N_\mathrm{f})$ by gauging an $\mathrm{SU}(N_\mathrm{c})$ color subgroup is equivalent to a conformally invariant $\mathrm{SU}(N_\mathrm{c})$ WZ model with level $k=N_\mathrm{f}+2N_\mathrm{c}$. The results of Knizhnik and Zamolodchikov [12] yield a value of the conformal anomaly c for the WZ model:

$$c=kD/(k+C_\mathrm{v})\,, \tag{10}$$

where D is the dimension of the color group, k is the level and C_v is the quadratic Casimir in the adjoint representation of $\mathrm{SU}(N_\mathrm{c})$. For our problem we find

$$c=\frac{(N_\mathrm{f}+2N_\mathrm{c})(N_\mathrm{c}^2-1)}{N_\mathrm{f}+3N_\mathrm{c}}+1+N_\mathrm{f}N_\mathrm{c}>1\,, \tag{11}$$

where the second term comes from the U(1) sector and the third term from the free left-handed fermions.

Thus, a constrained Fermi theory is equivalent to a Wess–Zumino sigma model on the color subgroup with nontrivial values of k and c. Note however that this is *not* a coset space σ-model.

It is instructive to consider a constraint acting directly on components of the energy–momentum tensor. Indeed, Goddard, Kent and Olive [8] showed that the difference of two properly normalized Virasoro algebras form a Virasoro algebra with a value of c equal to the difference of c for each of the terms.

Consider for instance a theory with N_c colors and N_f flavors of left-handed free fermions. The symmetry group of this ($k=1$) free theory is $\mathrm{SU}(N_\mathrm{c}N_\mathrm{f})$ (after having discarded a trivial U(1) piece). We now impose the constraint that the contribution of the first (say) p color components to the energy–momentum tensor are set to zero (rather than the color currents) by introducing the factor $\prod_{i=1}^p\delta(\psi^{if\dagger}\partial_+\psi^{if})$ in the

[#1] Since we are dealing with fermions in a complex representation, there is an additional U(1) symmetry which plays no role in the present discussion.

free-fermion generating functional. The Virasoro generators of the constrained theory are then trivially equal to the difference of the Virasoro generators of the unconstrained theory and those of the "deleted subspace". Thus the value of c is also the difference. More specifically c for the unconstrained theory is $N_c N_f$ and we are subtracting p modes. Then the constrained theory has

$$c = N_c N_f - pN_f = (N_c - p) N_f \,. \qquad (12)$$

The symmetry group of the remaining (free) theory is $SU((N_c - p)N_f)$. It is then clear that the physical content of both types of constraints is quite different. The problem of finding a field theoretic description of the $c<1$ FQS principal series remains open.

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References

[1] L. Onsager, Phys. Rev. 65 (1944) 117.
[2] T.D. Shultz, D.C. Mattis and E.H. Lieb, Rev. Mod. Phys. 36 (1964) 856.
[3] A.A. Belavin, A.M. Polyakov and A.B. Zamolodchikov, J. Stat. Phys. 34 (1984) 763; Nucl. Phys. B 241 (1984) 333.
[4] D. Friedan, Z. Qiu and S. Shenker, Phys. Rev. Lett. 52 (1984) 1575; in: Vertex operators in mathematics and physics, eds. J. Lepowsky et al. (Springer, Berlin, 1984).
[5] J.L. Cardy, in: Phase transitions and critical phenomena, Vol. 11, eds., C. Domb and M.S. Green (Academic Press, New York, 1986).
[6] E. Witten, Commun. Math. Phys. 92 (1984) 455.
[7] A.M. Polyakov and P.B. Wiegmann, Phys. Lett. B 131 (1983) 121; B 141 (1984) 223.
[8] P. Goddard, A. Kent and K. Olive, Phys. Lett. B 152 (1985) 88; Commun. Math. Phys. 103 (1986) 105.
[9] A. Kent, Phys. Lett. B 173 (1986) 403.
[10] L. Alvarez-Gaumé and P. Ginsparg, Nucl. Phys. B 243 (1984) 449.
[11] E. Fradkin, C.M. Naón and F.A. Schaposnik, to be published.
[12] V.G. Knizhnik and A.B. Zamolodchikov, Nucl. Phys. B 247 (1984) 83.